COLLEGE / UNIVERSITY

Department of Information Technology / Computer Science

B.TECH — LECTURE NOTES

FUNDAMENTALS OF QUANTUM COMPUTING

UNIT I: Foundations of Quantum Mechanics for Computing

Course Unit: 300 minutes (5 hours)

Prepared in the style of the supplied sample lecture-notes document

Unit I — Syllabus Coverage

This unit establishes the mathematical and conceptual foundation required to understand quantum computation. It introduces the quantum state, qubit, superposition, phase, interference, Bloch-sphere representation, measurement, multi-qubit states, and entanglement, followed by a first hands-on session with a quantum simulator.

Topics Covered

- Course introduction and why quantum computing matters
- Mathematical toolkit: complex numbers, vectors, matrices, unitary and Hermitian matrices, tensor products, and Hilbert space
- Classical bits versus qubits and Dirac notation
- Superposition, probability amplitudes, normalization, phase, and interference
- Bloch-sphere representation of a single qubit
- Measurement postulates, Born rule, collapse, repeated shots, and no-cloning
- Multi-qubit systems, tensor products, Bell states, and entanglement
- Hands-on introduction to a quantum simulator using Qiskit-style workflows
- Unit recap and self-assessment

Unit Learning Outcomes

1. Explain why quantum computing is studied and distinguish realistic capabilities from common hype.
2. Use basic complex-number, vector, and matrix notation required for quantum states and operations.
3. Represent a qubit using Dirac notation and explain the computational basis.
4. Calculate probabilities from quantum amplitudes using the Born rule.
5. Explain superposition, relative phase, and interference using simple examples.
6. Represent pure single-qubit states on the Bloch sphere.
7. Describe quantum measurement and explain why measurement statistics require repeated shots.
8. Construct two-qubit states using tensor products and distinguish product states from entangled states.
9. Describe the Bell states and the role of entanglement as a computational resource.
10. Create and inspect simple circuits on a quantum simulator and interpret state vectors and measurement histograms.

Part A: Course Introduction and Mathematical Toolkit

A.1 Introduction — Why Quantum Computing Matters

Quantum computing is a model of computation based on quantum-mechanical states and transformations. Instead of representing information only with classical bits, a quantum computer manipulates quantum states whose amplitudes can exhibit superposition and interference, and whose multi-qubit states can exhibit entanglement.

The goal is not to replace classical computers for every task. Quantum computing is being developed because certain computational problems have structures that can potentially be exploited by quantum algorithms. Examples discussed in this course include integer factoring, unstructured search, quantum simulation, and selected optimization or machine-learning problems.

A.1.1 Classical Limits and Quantum Motivation

- Cryptography: Shor's algorithm provides a theoretical polynomial-time approach to factoring and related period-finding problems on a sufficiently large fault-tolerant quantum computer.
- Search: Grover's algorithm provides a quadratic query improvement for unstructured search.
- Chemistry and materials: quantum systems can be simulated using quantum states, motivating applications in molecular and materials science.
- Optimization and machine learning: quantum algorithms and hybrid methods are active research areas, but practical advantage depends strongly on the problem, hardware, and algorithm.

A central theme of the course is hype versus reality. Present-day quantum processors are noisy and limited. A useful learning habit is therefore to separate a mathematical speedup, a laboratory demonstration, and a practical advantage in a real application.

A.1.2 Course Assumptions

- Basic linear algebra: vectors, matrices, matrix multiplication, and complex numbers.
- Basic discrete mathematics and probability.
- No prior quantum mechanics is required; the unit introduces only the quantum concepts needed for computing.

A.2 Complex Numbers

A complex number has the form $z = a + ib$, where a and b are real numbers and $i^2 = -1$. Quantum amplitudes are generally complex, so complex arithmetic is part of the basic mathematical toolkit.

- Real part: $\mathrm{Re}(z) = a$
- Imaginary part: $\mathrm{Im}(z) = b$
- Complex conjugate: $z^* = a - ib$
- Magnitude: $|z| = \sqrt{a^2 + b^2}$
- Squared magnitude: $|z|^2 = a^2 + b^2$

Euler's formula, $e^{i\varphi} = \cos\varphi + i \sin\varphi$, provides a convenient connection between complex phase and geometry in the plane.

Worked Example A1: Complex Amplitude

Let $\alpha = (1 + i)/2$. Its squared magnitude is $|\alpha|^2 = (1^2 + 1^2)/4 = 1/2$. Thus, if α is an amplitude associated with an outcome, its contribution to the corresponding probability is 1/2.

A.3 Vectors, Inner Products, and Normalization

A qubit state is represented by a normalized complex vector. In column-vector notation, a general single-qubit state is $|\psi\rangle = [\ \alpha\ ;\ \beta\ ]$, where α and β are complex amplitudes and $|\alpha|^2 + |\beta|^2 = 1$.

- Inner product: $\langle\varphi|\psi\rangle$
- Norm: $||\psi|| = \sqrt{(\langle\psi|\psi\rangle)}$
- Normalized state: $\langle\psi|\psi\rangle = 1$

The normalization condition ensures that the total probability of all measurement outcomes is one.

A.4 Dirac (Bra-Ket) Notation

- $|\psi\rangle$ is a ket representing a state vector.
- $\langle\psi|$ is the corresponding bra, obtained using the conjugate transpose.
- $\langle\varphi|\psi\rangle$ is an inner product and is a complex number.
- $|\psi\rangle\langle\varphi|$ is an outer product and produces a matrix.

The computational basis for one qubit consists of $|0\rangle = [1,0]^T$ and $|1\rangle = [0,1]^T$. Every single-qubit state can be expressed as $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$.

A.5 Matrices, Unitary and Hermitian Operators

Quantum operations are represented by matrices. A matrix U is unitary when $U^\dagger U = I$, where $U^\dagger$ is the conjugate transpose. Unitary evolution preserves normalization and therefore preserves total probability.

A Hermitian matrix satisfies $A^\dagger = A$. Hermitian operators are important for observables and measurement theory because their eigenvalues are real.

- Matrix multiplication represents composition of operations.
- Conjugate transpose is obtained by transposing and complex-conjugating each element.
- Identity matrix I leaves a state unchanged.
- Unitary matrices preserve inner products and norms.

A.6 Tensor Product and Hilbert Space

The tensor product combines the state spaces of separate quantum systems. Two qubits therefore occupy a four-dimensional complex state space with basis states $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$.

More generally, n qubits require a 2^n-dimensional state vector. This exponential growth is one reason quantum-state simulation becomes difficult on classical computers.

A Hilbert space is the mathematical vector space in which quantum states are represented. For introductory quantum computing, it is useful to think of it as the complete mathematical stage on which states, amplitudes, and operators are defined.

Part B: Classical Bits, Qubits, and Superposition

B.1 Classical Bit versus Qubit

A classical bit takes one of two logical values, 0 or 1. A qubit is a normalized quantum state in a two-dimensional complex vector space and can be written as $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$.

B.1.1 Computational and Alternative Bases

- Z/computational basis: $|0\rangle$ and $|1\rangle$
- X basis: $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ and $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$
- Y basis: $|+i\rangle = (|0\rangle + i|1\rangle)/\sqrt{2}$ and $|-i\rangle = (|0\rangle - i|1\rangle)/\sqrt{2}$

Different bases provide different ways of describing and measuring the same quantum system. Basis choice becomes important when discussing phase, interference, and measurement.

B.2 Physical Realizations of Qubits

- Superconducting qubits: engineered circuits operated at very low temperatures.
- Trapped-ion qubits: quantum states encoded in internal states of trapped ions.
- Photonic qubits: information encoded in properties of photons.
- Spin-based qubits: information encoded in quantum states associated with spin.

These platforms differ in physical implementation, control methods, connectivity, noise characteristics, and engineering constraints. The abstract qubit model lets algorithms be discussed independently of a particular hardware platform.

B.3 Superposition

Superposition means that a quantum state can be a linear combination of basis states. For a single qubit, $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$.

The statement that a qubit is 'both 0 and 1 at once' is a simplified slogan and can be misleading. What is mathematically true is that before measurement the state can contain amplitudes for multiple basis outcomes. Measurement produces an outcome according to the relevant probability distribution.

B.3.1 Probability Amplitudes

- Amplitude α corresponds to outcome 0 in the computational basis.
- Amplitude β corresponds to outcome 1.
- Probability of 0 = $|\alpha|^2$.
- Probability of 1 = $|\beta|^2$.
- Probabilities sum to one because the state is normalized.

B.4 Global Phase and Relative Phase

Multiplying an entire state by the same phase factor e^{iγ} produces a global phase. A global phase cannot be observed in ordinary measurement probabilities. By contrast, the relative phase between components can affect interference and therefore can change observable outcomes after later quantum operations.

Worked Example B1: Equal Superposition

For $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, both amplitudes have magnitude $1/\sqrt{2}$. Measuring in the computational basis gives $P(0) = 1/2$ and $P(1) = 1/2$.

Worked Example B2: Same Probabilities, Different Relative Phase

Consider $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ and $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$. Direct computational-basis measurement gives 50% for each outcome in both cases, but the relative phase differs. A later interference operation can distinguish the two states.

B.5 Interference

Quantum amplitudes, rather than probabilities, are combined during state evolution. Amplitudes can reinforce or cancel one another. This constructive and destructive interference is a central mechanism behind quantum algorithms.

- Constructive interference increases the amplitude of selected outcomes.
- Destructive interference reduces or cancels amplitudes of other outcomes.
- Useful quantum algorithms are designed so that interference amplifies desired answers and suppresses unwanted ones.

Part C: Bloch Sphere Representation

C.1 From Amplitudes to Geometry

Any pure single-qubit state can be written, up to an overall global phase, as $|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\varphi} \sin(\theta/2)|1\rangle$. The two real parameters θ and φ identify a point on the surface of the Bloch sphere.

C.1.1 Geometry

- North pole: $|0\rangle$
- South pole: $|1\rangle$
- Equator: equal-magnitude superpositions.
- Azimuthal angle φ represents relative phase.
- Polar angle θ determines the balance between $|0\rangle$ and $|1\rangle$.

The Bloch sphere is a visualization of a single qubit. It does not provide a direct geometric picture of an arbitrary multi-qubit state.

C.2 Important States on the Bloch Sphere

- $|0\rangle$ and $|1\rangle$ lie at opposite poles.
- $|+\rangle$ and $|-\rangle$ lie on opposite points of the x-axis.
- $|+i\rangle$ and $|-i\rangle$ lie on opposite points of the y-axis.

Quantum gates introduced in later units can be interpreted geometrically as rotations or combinations of rotations on the Bloch sphere.

Worked Example C1: Bloch Coordinates

For the state $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, $\theta = \pi/2$ and $\varphi = 0$, so the state lies on the positive x-axis of the Bloch sphere.

For $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$, $\theta = \pi/2$ and $\varphi = \pi$, corresponding to the opposite x-axis direction.

C.3 Pure and Mixed States

Pure single-qubit states correspond to points on the surface of the Bloch sphere. Mixed states are statistical mixtures and are represented by points inside the sphere. Mixed states are introduced here only conceptually; density matrices are treated more fully when noise is discussed.

C.4 Visualization and Simulation

A quantum simulator can display a state vector and, for a single qubit, a Bloch-sphere representation. Such visualizations are useful for connecting algebraic expressions to geometric intuition.

Part D: Measurement Postulates

D.1 Measurement and the Born Rule

Measurement connects the mathematical quantum state to classical outcomes. For a computational-basis measurement of $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, the Born rule gives $P(0) = |\alpha|^2$ and $P(1) = |\beta|^2$.

A measurement produces one outcome, not a list of all amplitudes. The amplitudes determine the probability distribution from which the observed result is sampled.

D.2 State Update and Collapse

In the ideal projective-measurement picture used in this course, after measuring in the computational basis, the state is projected onto the observed basis state. If the result is 0, the post-measurement state is $|0\rangle$; if the result is 1, it is $|1\rangle$.

This means that repeatedly measuring the same already-collapsed state in the same basis produces the same result, whereas repeated independent preparations of a superposition produce a distribution of outcomes.

D.3 Shots and Histograms

Quantum hardware and simulators commonly repeat a circuit many times. Each repetition is called a shot. The resulting counts can be displayed as a histogram. As the number of shots increases, the empirical frequencies generally approach the probabilities predicted by the quantum state.

Worked Example D1: 1024 Shots

Suppose a qubit is prepared in $|+\rangle$ and measured 1024 times in the computational basis. The ideal probabilities are 0.5 and 0.5. An actual finite-shot result might be close to 512 counts for 0 and 512 counts for 1, but exact equality is not guaranteed.

D.4 Measuring in Different Bases

Measurement is basis-dependent. A state that is deterministic in one basis can be a superposition in another basis. For example, $|+\rangle$ is a definite +1 eigenstate of the X-basis measurement but produces equal probabilities for 0 and 1 under computational-basis measurement.

D.5 No-Cloning Theorem

The no-cloning theorem states that an unknown arbitrary quantum state cannot be copied perfectly by a physical operation that works for every possible input state. This is different from ordinary classical copying and is an important property of quantum information.

No-cloning does not mean that quantum information can never be reproduced in any form. Known basis states can be prepared again, and quantum error-correction schemes use encoded redundancy without making forbidden copies of an unknown state.

Part E: Multi-Qubit Systems and Entanglement

E.1 Building Multi-Qubit States

For two qubits, the tensor product of the individual state spaces produces four computational-basis states: |00⟩, |01⟩, |10⟩, and |11⟩. A general two-qubit pure state can therefore be written as a linear combination of these four basis states.

For n qubits, the computational basis contains 2^n basis states. This exponential state-space dimension is central to quantum computing, although it should not be interpreted as 2^n independently readable classical values.

E.2 Product States

If two qubits can be written as $|\psi\rangle \otimes |\varphi\rangle$, their joint state is a product or separable state. For example, $|+\rangle \otimes |0\rangle$ is separable.

Worked Example E1: Tensor Product

Let $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. Then $|+\rangle \otimes |0\rangle = (|00\rangle + |10\rangle)/\sqrt{2}$. Measuring in the computational basis gives 00 and 10 with equal probability.

E.3 Entanglement

An entangled state is a joint quantum state that cannot be expressed as a tensor product of independent single-qubit states. Entanglement creates correlations that cannot be represented by assigning each qubit an independent pure state.

E.3.1 Bell States

- $|\Phi+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$
- $|\Phi-\rangle = (|00\rangle - |11\rangle)/\sqrt{2}$
- $|\Psi+\rangle = (|01\rangle + |10\rangle)/\sqrt{2}$
- $|\Psi-\rangle = (|01\rangle - |10\rangle)/\sqrt{2}$

The Bell states are maximally entangled two-qubit states. For example, measuring the first qubit of $|\Phi+\rangle$ in the computational basis gives 0 or 1 with equal probability, and the second qubit is then found in the corresponding same value.

E.4 GHZ and W States — Brief Introduction

For more than two qubits, different forms of multipartite entanglement occur. A three-qubit GHZ state has the form $(|000\rangle + |111\rangle)/\sqrt{2}$. A W state has the form $(|001\rangle + |010\rangle + |100\rangle)/\sqrt{3}$. These examples illustrate that multipartite entanglement has structures richer than a simple pairwise correlation.

E.5 EPR and Bell Inequality — Conceptual View

Entanglement motivated foundational questions about whether quantum correlations could be explained by local hidden variables. Bell inequalities provide experimentally testable constraints that are violated by quantum predictions for suitable entangled states and measurements. The purpose here is conceptual: the result demonstrates that quantum correlations have properties not reproduced by a broad class of classical local hidden-variable models.

E.6 Entanglement as a Computational Resource

- Quantum teleportation uses shared entanglement together with classical communication to transfer an unknown quantum state.

- Superdense coding uses shared entanglement to communicate two classical bits using the transmission of one qubit, under the protocol's assumptions.
- Many quantum algorithms and protocols exploit entanglement together with superposition and interference.

Part F: Hands-On Session — First Steps with a Quantum Simulator

F.1 Simulator Workflow

The first practical session connects the mathematical concepts to executable circuits. The workflow is: create a quantum register → initialize or apply operations → inspect the state → measure → repeat for multiple shots → interpret the histogram.

- Access a quantum-computing environment such as Qiskit and a simulator.
- Create a one- or two-qubit circuit.
- Apply simple gates introduced conceptually in this unit.
- Inspect the state vector when the simulator supports it.
- Plot or inspect a Bloch-sphere representation for a single qubit.
- Run repeated shots and interpret the measurement counts.
- Create and measure a Bell pair.

F.2 Guided Exercise: Equal Superposition

11. Start with a qubit in $|0\rangle$.
12. Apply a Hadamard operation to create $|+\rangle$.
13. Inspect the state vector and confirm approximately equal amplitudes for $|0\rangle$ and $|1\rangle$.
14. Measure repeatedly.
15. Compare the histogram with the theoretical probabilities 0.5 and 0.5.

The key learning point is that a state vector describes amplitudes, while a measurement histogram describes sampled classical outcomes.

F.3 Guided Exercise: Bell Pair

16. Prepare two qubits in $|00\rangle$.
17. Create a superposition on the first qubit.
18. Apply a controlled operation so that the two qubits become entangled.
19. Measure both qubits repeatedly.

For the standard Bell state $|\Phi+\rangle$, the expected computational-basis outcomes are 00 and 11, each with probability 1/2. The important observation is correlation: the results are not independently random.

F.4 AI-Assisted Linear Algebra Practice

AI tools may be used as study assistants for checking matrix multiplication, normalization, inner products, and tensor products. Students should verify each result independently rather than treating an AI-generated calculation as authoritative.

- Ask for a worked example of normalizing a two-component complex vector.
- Generate a similar practice problem and solve it independently.
- Use the tool to explain a mistake after attempting the problem yourself.
- Compare an AI-generated answer against direct calculation or a simulator.

Part G: Worked Examples and Integrated Problems

Worked Example 1: Normalize a Qubit

Given $|\psi\rangle = a|0\rangle + b|1\rangle$ with $a = 1/\sqrt{3}$ and $b = \sqrt{(2/3)}$, verify normalization: $|a|^2 + |b|^2 = 1/3 + 2/3 = 1$. Therefore the state is normalized.

Worked Example 2: Measurement Probabilities

For $|\psi\rangle = (\sqrt{3}/2)|0\rangle + (1/2)|1\rangle$, $P(0) = 3/4$ and $P(1) = 1/4$. Over many shots, approximately three quarters of the results should be 0 and one quarter should be 1, subject to statistical fluctuations.

Worked Example 3: A Relative-Phase Experiment

Prepare either $|+\rangle$ or $|-\rangle$. Both give identical 50/50 statistics in the computational basis. Apply a suitable interference operation before measurement. The two states can then produce different deterministic outcomes. This demonstrates that relative phase can be physically meaningful even when it is invisible in a direct computational-basis probability measurement.

Worked Example 4: Product versus Entangled State

Consider $|\psi 1\rangle = (|00\rangle + |01\rangle)/\sqrt{2}$ and $|\psi 2\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$. The first can be factored as $|0\rangle \otimes |+\rangle$ and is therefore separable. The second is a Bell state and cannot be factored into one-qubit states, so it is entangled.

Integrated Exercise

A student prepares two qubits, applies a superposition operation to the first qubit, and then entangles the pair. The student obtains approximately equal numbers of 00 and 11 outcomes over many shots.

20. Identify a plausible target state.
21. Write its ket representation.
22. State whether the pair is separable or entangled.
23. Explain why the individual results can look random while the pair remains correlated.
24. Explain what additional information a state-vector simulator provides beyond the histogram.

Unit Summary

- Quantum computing uses quantum states and unitary transformations to process information; it is not simply a faster version of classical computing for every problem.
- Complex numbers, vectors, matrices, inner products, normalization, tensor products, and Hilbert spaces form the mathematical toolkit used throughout quantum computing.
- A qubit is represented as $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$.
- Measurement probabilities are determined by the squared magnitudes of the relevant amplitudes through the Born rule.
- Superposition describes linear combinations of basis states; relative phase matters because it affects interference.
- The Bloch sphere provides a geometric representation of pure single-qubit states.
- Measurement converts quantum information into classical outcomes and is intrinsically probabilistic for a general superposition.
- The no-cloning theorem prevents perfect copying of an arbitrary unknown quantum state.
- Tensor products describe multi-qubit systems, whose state-space dimension grows as 2^n.
- Entangled states cannot be decomposed into independent single-qubit states; Bell states provide the standard two-qubit examples.
- Quantum simulators allow students to inspect state vectors, visualize single-qubit states, run shots, and compare experimental statistics with theoretical probabilities.

Review Questions

25. Why is quantum computing not expected to outperform classical computing on every problem?
26. Define a complex number and explain why complex numbers are needed for quantum amplitudes.
27. What is the normalization condition for a single-qubit state?
28. Explain bra-ket notation and distinguish $|\psi\rangle$ from $\langle\psi|$.
29. What is a unitary matrix and why is unitarity important in quantum computation?
30. What is the tensor product and why is it required for multi-qubit systems?
31. Differentiate a classical bit from a qubit.
32. Define superposition and explain why the phrase 'a qubit is both 0 and 1 at once' can be misleading.
33. Calculate the measurement probabilities for $|\psi\rangle = (\sqrt{3}/2)|0\rangle + (1/2)|1\rangle$.
34. What is the difference between global phase and relative phase?
35. Explain constructive and destructive interference of quantum amplitudes.
36. What information is represented by the polar and azimuthal angles on the Bloch sphere?
37. Distinguish pure states on the Bloch-sphere surface from mixed states inside the sphere.
38. State the Born rule for computational-basis measurement.
39. What is a measurement shot and why are many shots useful?
40. State the no-cloning theorem in your own words.
41. What is the difference between a product state and an entangled state?
42. Write the four Bell states.
43. What happens when one qubit of $|\Phi+\rangle$ is measured in the computational basis?
44. Why does the dimension of an n-qubit state vector scale as 2^n?
45. Describe one way entanglement is used as a computational or communication resource.
46. Explain the difference between a state vector and a measurement histogram.
47. Design a simple simulator experiment that demonstrates superposition.
48. Design a simple simulator experiment that demonstrates entanglement.

Self-Assessment Checklist

- I can normalize a single-qubit state.
- I can calculate measurement probabilities from amplitudes.
- I can explain global and relative phase.
- I can identify the standard single-qubit basis states.
- I can locate $|0\rangle$, $|1\rangle$, $|+\rangle$, $|-\rangle$, $|+i\rangle$, and $|-i\rangle$ conceptually on the Bloch sphere.
- I can form a two-qubit state using a tensor product.
- I can distinguish separable and entangled states.
- I can explain the correlation in a Bell-state measurement.
- I can run a simple quantum circuit on a simulator and interpret its histogram.

Suggested Mini-Lab

Build a three-part notebook: (1) prepare $|0\rangle$ and $|1\rangle$ and verify deterministic measurement; (2) prepare $|+\rangle$ and compare 128, 1024, and 4096 shots; (3) prepare a Bell state and compare individual-qubit statistics with joint two-qubit statistics. Record the theoretical probabilities, simulator results, and a short interpretation of any finite-shot differences.